

E-commerce Analytics on AWS

From Raw Transactions to Analytics Pipeline

Samarth Singh

Tools:- Python | AWS S3 | ATHENA
|Power BI



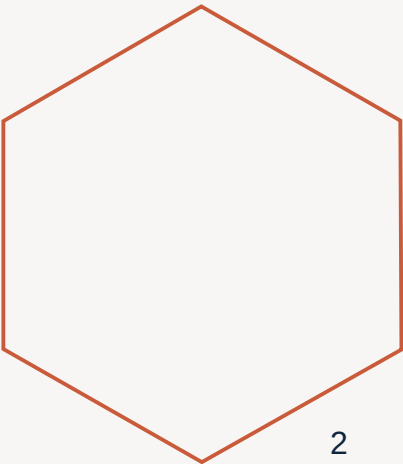
Problem Statement

Objective :-

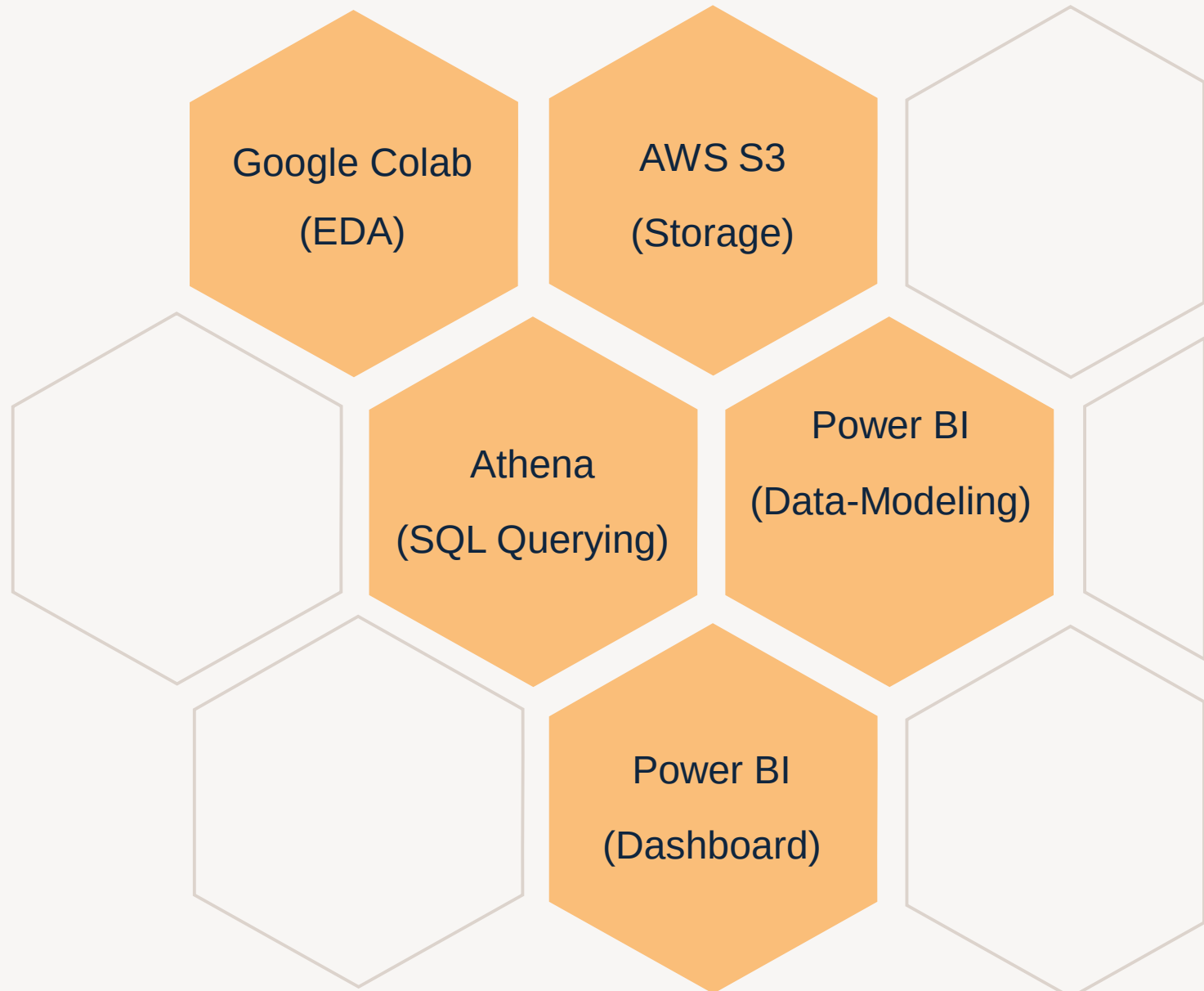
- Analyze e-commerce transaction data to uncover sales trends, customer behaviour, and Profitability drivers.
- Build a cloud-based pipeline for scalable, repeatable analysis.

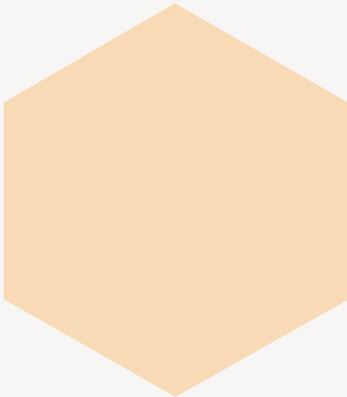
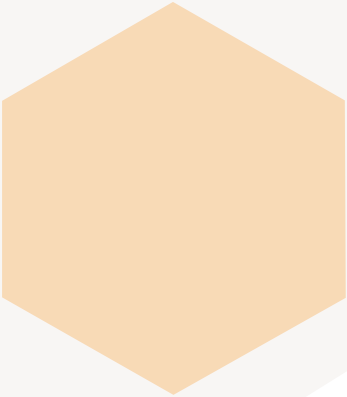
Dataset Overview

Orders	Customers	Products
Rows - 34,500 Columns – order_id, customer_id, product_id, order_date, quantity, discount, total_amount, profit, shipping_cost, delivery_time, region, returned	Rows - 7,903 Columns - customer_id, age, gender	Rows - 24912 Columns - product_id, category, price



Architecture



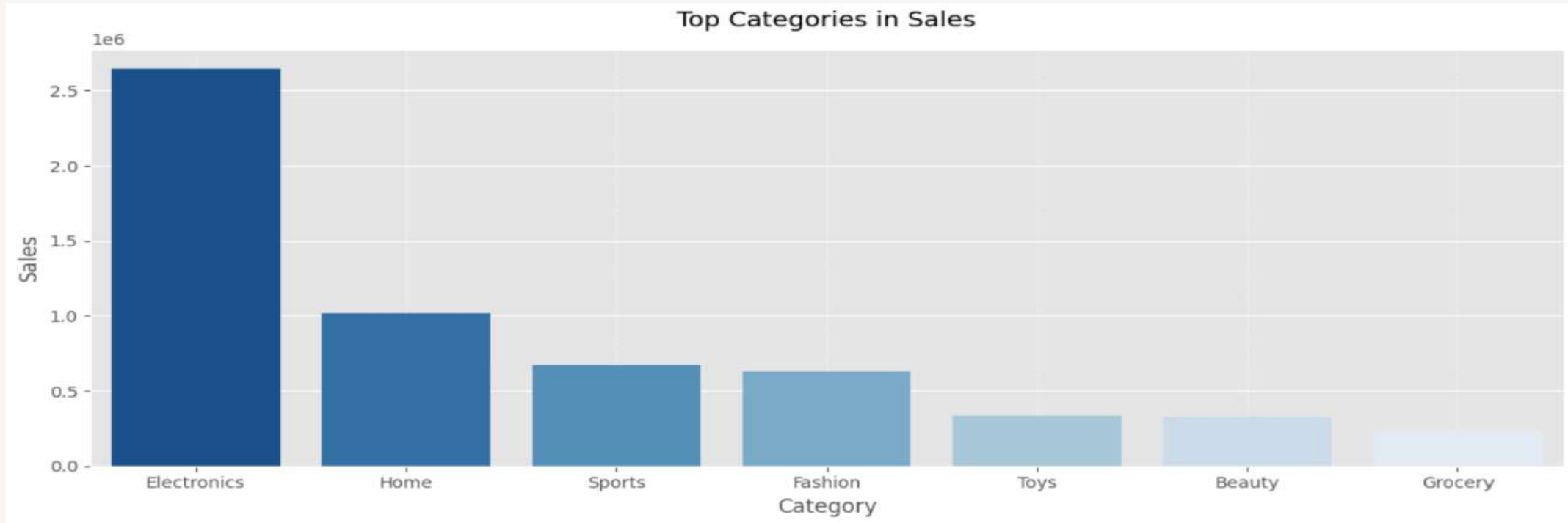


Data Quality & Cleaning

Issue Found :- Inconsistent customer/product attributes across duplicate IDs

Fix Applied :- 'Latest record wins' rule for data consistency

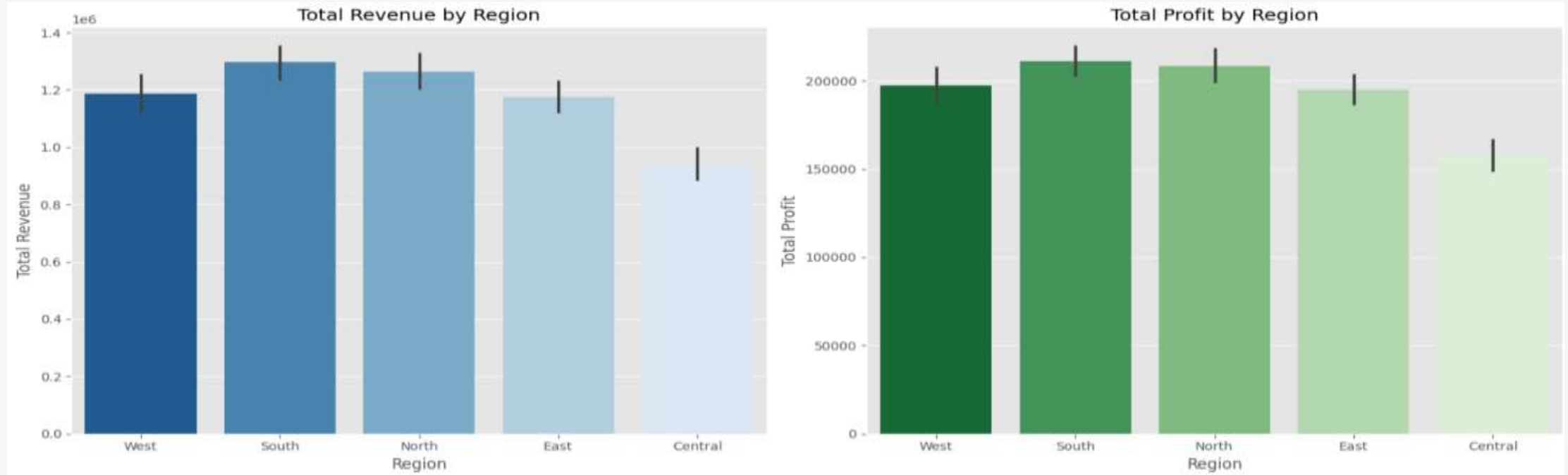
Revenue Concentration



Electronics = 56.6% of the total revenue

- Exceed combined shares of all other 6 categories
- Indicates category concentration risks

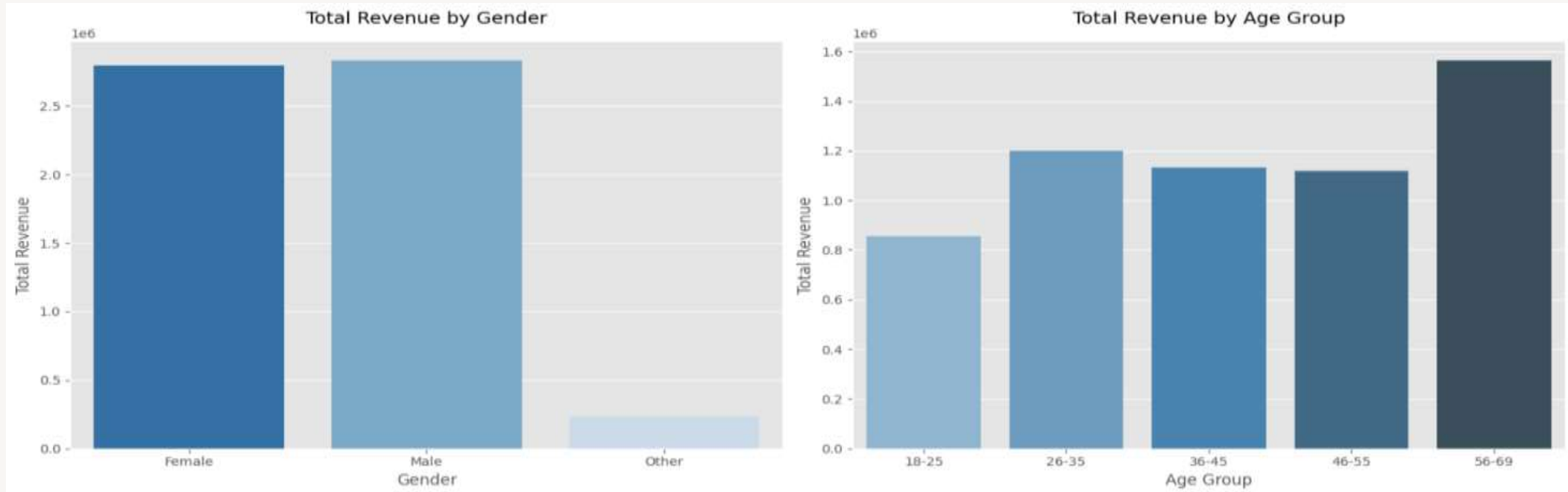
Regional Performance



Revenue & Profit by Region

- South: Highest revenue (\$1.3M) and Profit (\$211K)
- Central: lowest on both metrics
- Profit ranking mirrors revenue ranking → margins consists across regions

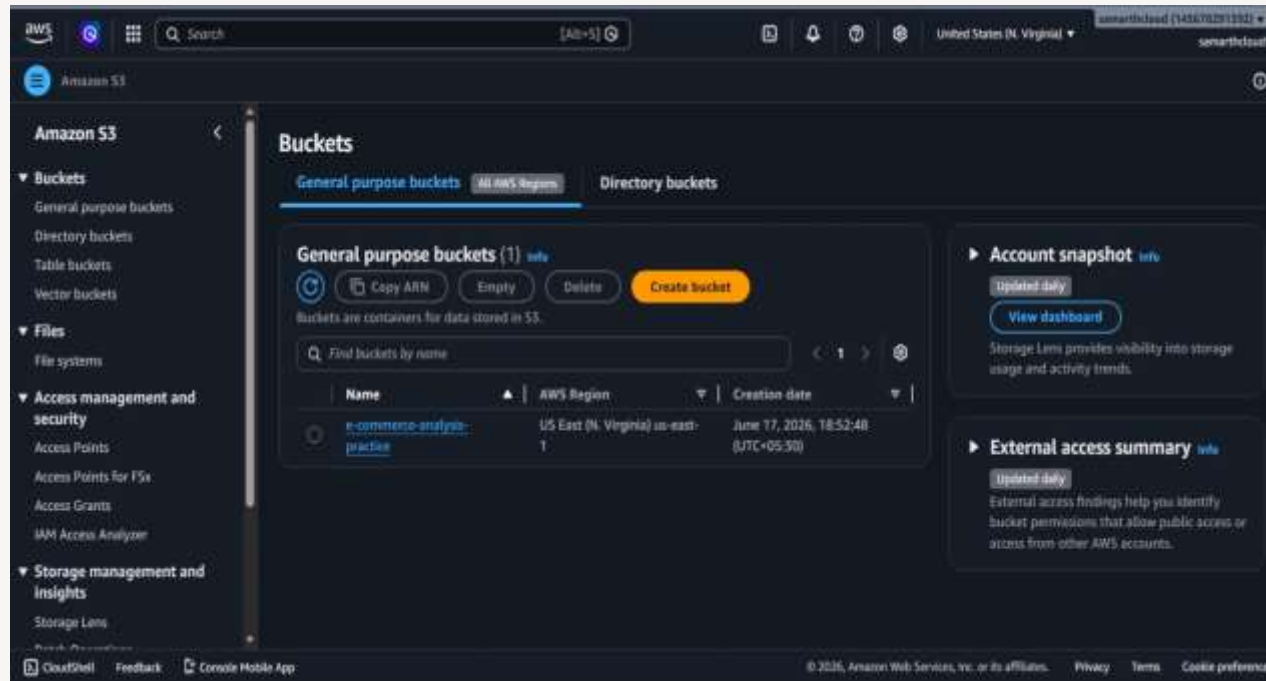
Customer Demographics



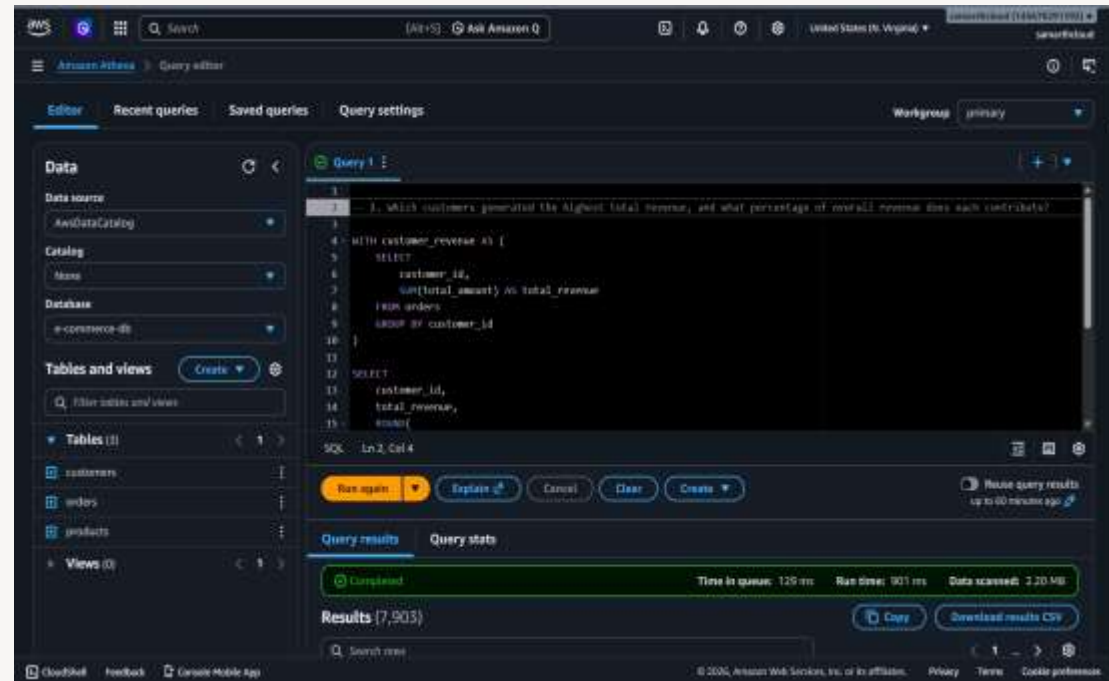
Revenue across gender & age-group

- 56-69 age-group: highest revenue contributor(\$1.5M)
- Gender split: Female \$2.8M vs Male \$2.7 (minimal difference)
- Revenue is broadly distributed across customer base (no single-customer dependency)

AWS S3 + Athena



AWS S3

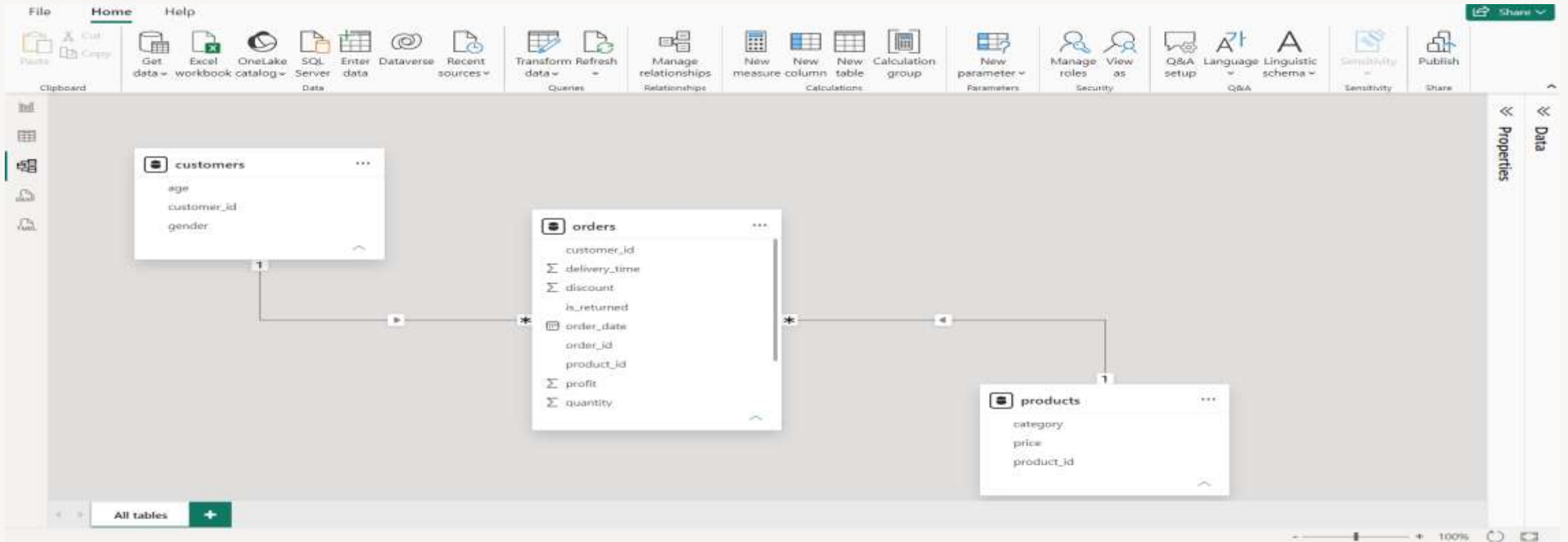


AWS Athena

S3 :- Centralized storage for cleaned transaction data

Athena :- Serverless SQL queries directly on S3 – no infrastructure, pay-per-query

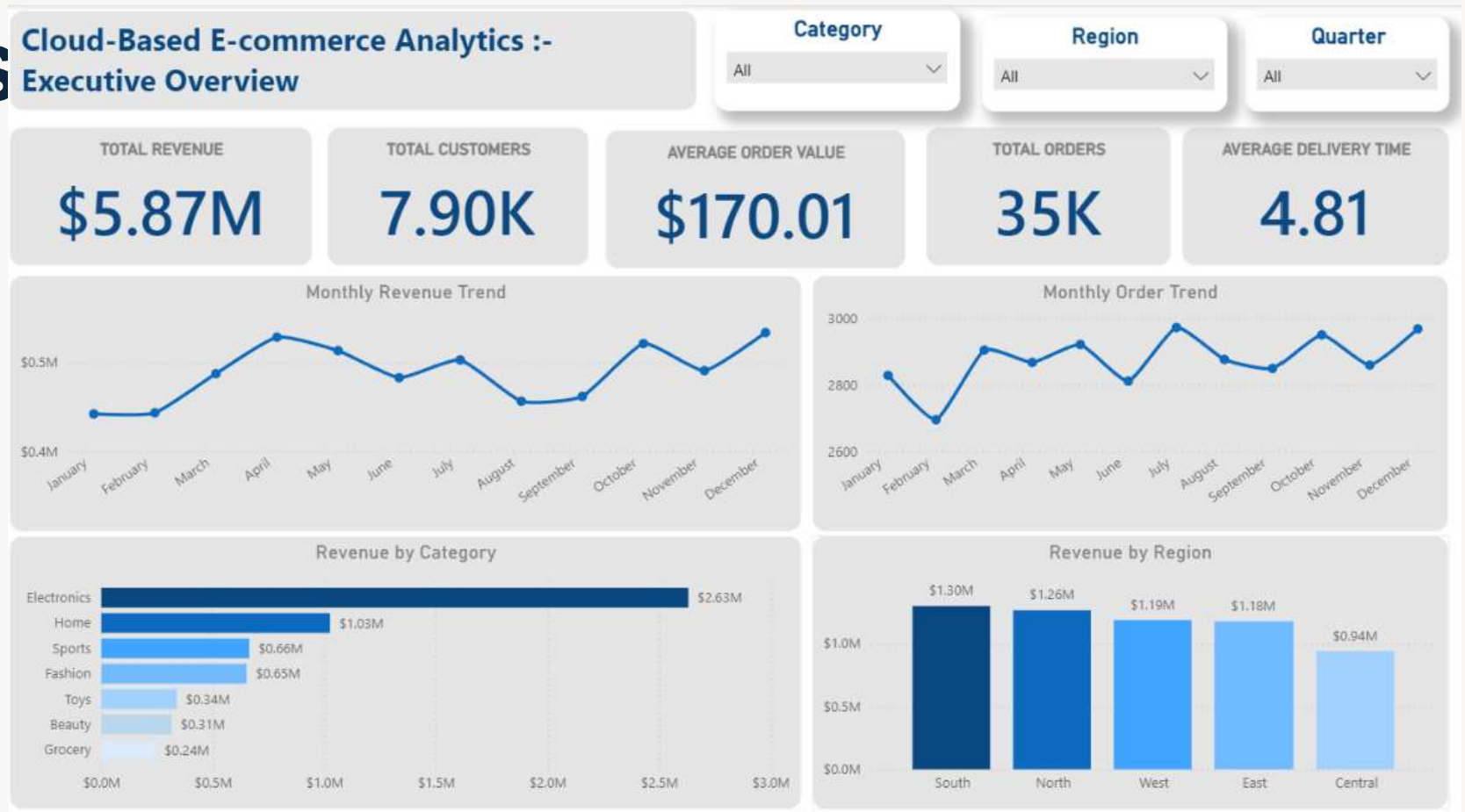
Power BI Data Modeling



Star Schema

Data import from S3 storage into Power BI
Star Schema – customer table → orders table ← product table

Key Business Insights



- Electronics drives over half of revenue → diversification risk
- Overall return rate: 5.52% | Fashion (8.3%) & Electronics (7.3%) return more than Grocery (1.3%)
- Price is the strongest driver of order value; delivery time show minimal correlation



Challenges & Learning

Challenge:

- Catching data integrity issue before analysis
- First experience with AWS

Learning:

- Cloud storage & serverless querying (S3, Athena)
- End to end pipeline thinking, not just isolated charts
- Validating data before visualizing it



Thank you

Samarth Singh

[E-mail- officialsamarthsingh@gmail.com](mailto:officialsamarthsingh@gmail.com)

[LinkedIn- https://www.linkedin.com/in/samarth-singh-062371215/](https://www.linkedin.com/in/samarth-singh-062371215/)

Git Hub: <https://github.com/OfficialSamarthsingh>

